Handover Report *25 September 2023*

Spectroscopic study of neutron-rich nuclei around mass $A = 104$

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1 Test Analysis

The singles spectra for mass $A = 106$ decay chain was analyzed to get used to peak finding and fitting. gf3 (RADWARE) was used for this analysis. The spectra plots have been saved in the below directory.
File: ~/uxb103/hgtot_analysis/spe_file/hgtot.spe

Next, the $\gamma - \gamma$ coincidence analysis was performed on the data from the ILL(06/23) experiment. The peak centroids(energy) were found using ROOT ShowPeaks() function.
File: ~/uxb103/hgtot_analysis/nb106−ill.root
The gated spectra for ^{106}Mo, ^{106}Nb, ^{106}Ru, ^{106}Tc have been saved in gg_plots sub-directory

2 ILL data

Processed file: /home/uxb103/ILL_06_23/157735_157760_processed_ion_coin.root

Singles : Singles analysis using gf3 was performed to identify all the gamma coming from the mass 104 decay chain. Energy calibration, ref Figure 1 has been performed to account for the offset between the ROOT exported .spe file and gf3.

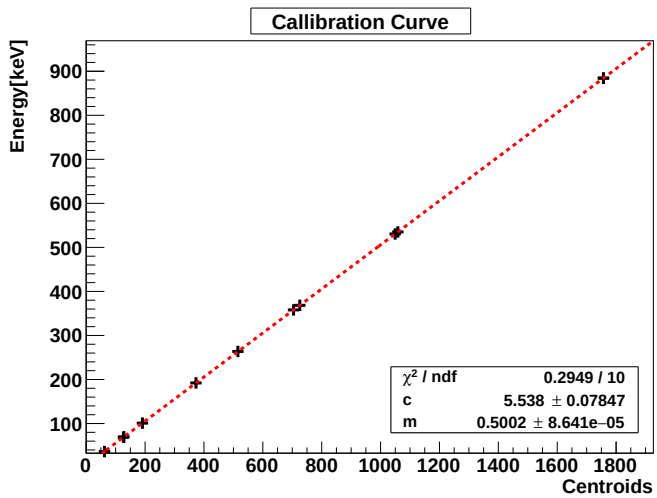

Figure 1: Calibration curve

Coincidence : Gamma gamma coincidence matrix was analysed by gating on ^{104}Nb transitions. The analysis was done in ROOT using /home/uxb103/aj−rtoolsill.cpp
Working directory: /home/uxb103/ILL_06_23/gamma_gamma_coin
Each gate(energy Ene) has a sub-directory G_Ene which contains the gated spectra images and fit parameters for all the gamma observed. The preliminary analysis revealed many Compton features and background.

A Compton filter(csub) was applied to remove the compton peaks and the background.
./StructEleBackSub nb106−all−2020−05−01.root hgg
A sanity check to ensure that the filter is not removing *too* many counts from real gamma was performed.

For this, counts under high-intensity, isolated, and directly fed peaks were compared with/without the filter. The test gave reasonably fair results.

2.1 Detection Efficiency

CARIBU data, being more reliable for relative yields, was used for extracting the gamma yields.
File: /home/uxb103/CARIBU/bgg_cs.esc The file was opened in xmesc, gated at the strongest transition of ^{104}Mo, 192, and exported to gf3. The yields (peak areas) were extracted using SUSBnf commands and compared for consistency. The extracted areas were corrected for CARIBU array efficiency to obtain the relative intensities.

$$I_\gamma = \frac{\#cts}{\epsilon}$$

The I_γ were compared with the ENSDF values, and there was a disagreement at some energies. The likely reason for this can be the presence of a beta decaying isomer in ^{104}Mo since the relative yields can vary with the ratio of metastable to ground state.

^{104}Ru, with no beta decaying isomer, was chosen for obtaining the yields. The extracted intensities agreed well with ENSDF values. Peak areas file: /home/uxb103/CARIBU/g_358keV/gf3.sto

These I_γ from CARIBU were used to find the detection efficiency of ILL.

$$\epsilon_\gamma = \frac{\#cts_{ILL}}{I_\gamma}$$

File for ILL counts and plots: /home/uxb103/ILL_06_23/compton_sub/det_eff

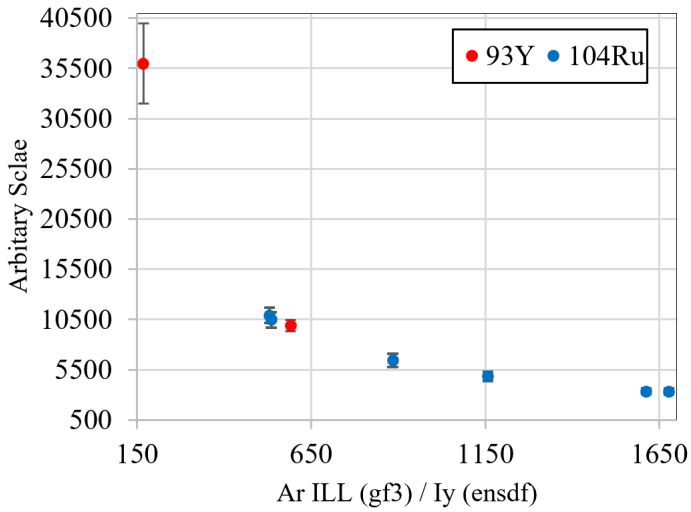

Figure 2: Detection Efficiency for ILL using ^{104}Ru and ^{93}Y (contaminant)

2.2 Sorting Setup

lst-data-process-128

The LstDataProcess library provides tools to read, write, and process lst binary files containing events with size 128 bits produced at the ILL. The tar archive currently resides at /data10tb−ajm/ILL−2023−06/232/Preparation

Listing 1: Compiling lst2root

```
$ tar -xvf lstdataprocess128-1.0.3.tar.gz
$ mkdir build && cd build
$ ../configure --prefix=/usr/local --enable-nomad=(yes|no) --enable-root=(
    yes|no) --enable-nappli= (yes|no)
$sudo make install
```

If there is a compilation error, it can usually be resolved by `const` → `constexpr` in file Coincidence-Offline.h
static const double _energyMax=10000

Listing 2: RunningLst2root128

```
$ Lst2root128|data location|time window in nanoseconds|clean time after
    window| multiplicity|config files|buffer size| outputfilelocation|
    verbose mode (t/f)|debugging (t/f)
```

Running above for a file run_no.evt will create a file /processed/run_no.root

Analysis_ConvElec2SiGePl-ajm.cpp

The code processes the root file for analysis, filling the various histograms and coincidence matrices. The relevant file and its /Makefile are currently located at
/data10tb−ajm/ILL−2023−06/232/Preparation/aj−**test**
Assuming these two files have been copied to a directory dir, the code can be compiled using command $make. An executable Analysis_dEE_ajm will be created

Listing 3: Executing Analysis_dEE_ajm

```
$ .\Analysis_dEE_ajm run_no_min  run_no_max
```

Executing this will create the final sorted file /analysed/run_no_processed.root

3 104 Nb level scheme

The same file [Section 2] has been used to build the level scheme by looking at $\gamma - \gamma$ coincidence relationships after applying the Compton filter to the 2D matrix(hgg_sub2).
Working Dir: /home/uxb103/ILL_06_23/compton_sub/level_scheme

Subdirectories `Gate_Ekev`- for a gate at energy E keV, contains the spectra plots with fits, fit_params.txt has the fit parameters from ROOT for all the peaks. `misc` - contains overlay spectra to verify further gamma that are in cascade. `print` - has the gated ranges for printing purposes.

Gates were applied at all the transitions of ^{104}Nb except for 8.7, 28.9, and 37.5 keV since they are in the Xray region.

A summary of the analysis is presented below. Further details about the $\gamma - \gamma$ analysis and the deduced relationships can be found in the scanned logbook and notes.

1. The 250.6 keV transition was not observed while gating 61.1 and 263.7 keV gamma.

2. A strong transition of 219 keV was observed while gating 213 keV and vice versa. However, gating on 241 and 140.9 keV no 219 keV was seen. Assuming the correctness of ENSDF branching ratios, this is not predicted since the three transitions(213, 241.9, and 140.9) depopulate the same levels. Most likely, this can be attributed to a contaminant.

3. 150.8 keV was observed in Gates at 392.6, 420.5, and 85.5 keV. Some feature at 150.2 keV was also seen in gated spectra of 100.9, 213, and 241.9 keV. One possibility for this is ^{104}Zr losing a neutron following the beta decay to ^{103}Nb, which has a transition of 105.6 keV. Secondly, ^{104}Mo also has a strong transition at 150.7 keV [1].

4. 392.6 keV was seen to be in coincidence with the 100.6, 150.8 keV transitions. The reciprocal coincidences were also observed. However, a gammas and level search revealed that ^{99}Zr has a transition of 392.7 keV, which can be a potential contaminant [2].

5. 420 keV,100.9, and 49.9 keV were observed to be coincident with each other. The reciprocal relations were true as well. Overlay plots were used to confirm these relations. Furthermore, a 49.9 keV peak was not observed while gating at 210 keV; there was just a single spike. This suggests that 420.5, 100.9, and 49.9 keV form a cascade relationship. The 2D $\gamma-\gamma$ matrix was also analyzed, ascertaining that 420 keV is not coming from Compton scattering. However, one possibility is contamination from ^{104}Tc transitions whose energies overlap with ^{104}Nb, 50, 101 keV, and are coincident with 421 keV.

6. 85.5, 100.9, and 150.2 keV were found to be coincident with each other. This relation requires further investigation. Additionally, 85.5 keV was found to be in coincidence with only 241.9 keV and not 213, 140.9 keV, while the latter three levels depopulate the same level(E = 250.6 keV), which is not physical. The possibility of Compton scattering contributing to 85.5 keV was ruled out by looking at the 2D coincidence matrix (Figure 3).

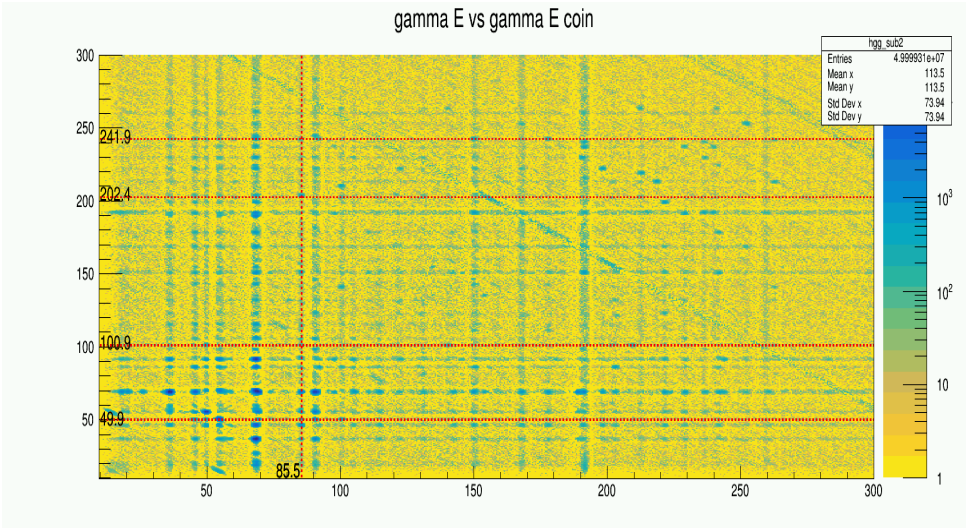

Figure 3: The 85.5 keV observed at gates of 49.9, 100.6, 202.4, and 241.9 is not due to Compton scattering as indicated by the absence of diagonal lines

7. 365.8 keV and 274.2 keV are coincident with each other. With this limited evidence, its placement is shown in Figure 4.

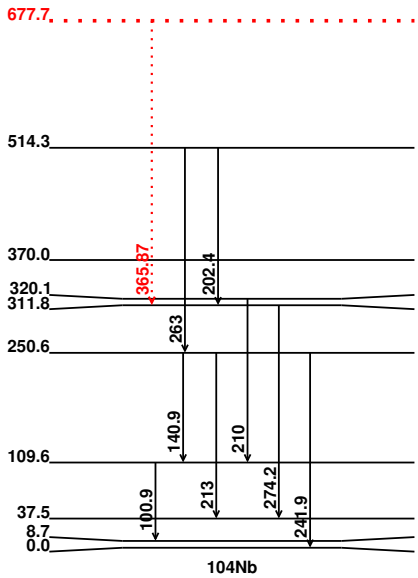

Figure 4: ^{104}Nb level scheme after incorporating the coincidence relationships

The 49.9 keV gamma, coincident with 100 keV, has not been placed in the level scheme since more investigation is needed to understand its coincidence relationships and confirm if 420.5 keV is a contaminant.

References

[1] Jean Blachot. Nucl. Data Sheets 108, 2035 (2007).

[2] BD Kern, K Sistemich, W-D Lauppe, and H Lawin. The beta decay of104nb and the level scheme of 104mo. *Zeitschrift für Physik A Atoms and Nuclei*, 306(2):161–170, 1982.